

code adds unused capacity while keeping one name per state. Finally, a code deliberately gives the same semantic state two possible names dependent on the path reached. The last pair isolates extra capacity from path-dependent naming. The main observation in Figure 12 is that canonical and padded codes recover all 16 states and remain exact through the tested horizon of 256. Mean final accuracy for the canonical interface, compared with 72.22% for one-pass reasoning, approaches 100% with context-bootstrap interval [12.50, 38.89]. The three-bit code has significant residual state uncertainty and never recovers the exact state. Extra capacity does not hurt the padded code, but path-dependent naming does: after three steps, the aliased code has only 30–40% exact state, versus 100% for the equally sized canonical code.